Stillness as a Controlled Stochastic Process

An Operational Model and Its Exact Causal Simulation

25 October 2026

Abstract

We formulate an operational model of stillness that distinguishes reduced mental content from reduced wakefulness or responsiveness. A latent-state model induces a finite controlled tree of experimental transcripts. Conditional on the finite causal Shannon theorem, this tree admits one finite seed and a shared-history-consistent decoder, exact for every deterministic adaptive experimental policy, within a universal additive entropy bound. We prove the reduction and observational non-identifiability under hidden extensions. These are mathematical consequences of explicit assumptions, not an established physiological theory. No claim of originality is made.

1 Operational variables

Fix a finite horizon n, a finite nonempty action set $\mathcal{A}$ and a finite nonempty outcome set $\mathcal{Y}$. An action may be a stimulus, a question, or a no-probe interval. An outcome encodes discretized physiological measurements, behavioral responses, and subjective reports. Missing responses must be recorded explicitly; intervention and recording schedules belong to the model.

Let S_t take values in a finite hidden-state set $\mathcal{S}$, with initial law ν. Assign latent scores

$$w, c, r : \mathcal{S} \longrightarrow [0, 1]$$

for wakefulness, intensity of mental content, and readiness to respond. These interpretations are modeling hypotheses, not directly measured facts.

Definition 1 (Candidate stillness region)**.** *For thresholds $w_0, r_0 \in (0, 1]$ and $c_0 \in [0, 1)$, put*

$$\mathcal{S}_{\text{still}} = \{s : w(s) \geq w_0,\ c(s) \leq c_0,\ r(s) \geq r_0\}.$$

Thresholds and their measurement calibration must be specified before testing. A report of mental silence alone does not identify this region.

Let K_a be stochastic transition matrices and O_a emission kernels. Our timing convention is

$$S_{t+1} \sim K_{A_t}(S_t, \cdot), \qquad Y_t \sim O_{A_t}(\cdot \mid S_{t+1}).$$

A probe may therefore change the state it measures. The controlled Markov and conditional emission structure are explicit assumptions.

2 Reduction to a controlled tree

Write $h = (a_0, y_0, \ldots, a_{t-1}, y_{t-1})$ for an observed history. Initialize $b_\varnothing = \nu$ and define

$$q_h^a(s') = \sum_s b_h(s) K_a(s, s'), \tag{1}$$

$$p(y \mid h, a) = \sum_{s'} q_h^a(s') O_a(y \mid s'), \tag{2}$$

$$b_{hay}(s') = \frac{q_h^a(s') O_a(y \mid s')}{p(y \mid h, a)}. \tag{3}$$

For application of the positive-kernel causal theorem, assume $p(y \mid h, a) > 0$ at every node of the finite tree. No extension to zero-probability branches is asserted here.

Proposition 1 (Exact transcript reduction)**.** *For every deterministic adaptive policy $A_t = \pi_t(H_t)$, the latent model and the tree with transition laws $p(\cdot \mid h, a)$ have the same transcript law:*

$$P_\pi(y_0, \ldots, y_{n-1}) = \prod_{t=0}^{n-1} p(y_t \mid h_t, \pi_t(h_t)).$$

Proof. Conditioned on h, the hidden-state law is b_h. Prediction through K_a and marginalization through O_a yield the displayed outcome law; Bayes' rule yields the posterior. Induction and the probability chain rule give the result. Deterministic actions contribute no additional conditional probability factors. □

3 The conditional causal Shannon bridge

An exact simulator uses a finite seed U, whose distribution is independent of the selected policy, and a single deterministic decoder

$$Y_t = D(U, H_t, A_t).$$

The same seed atom gives the same response at any shared history–action pair. Exactness is required for every deterministic adaptive policy. Let $C(T)$ be the infimum of $\mathrm{H}(U)$ over these finite simulators.

For each policy define the transcript information $I_\pi = -\log_2 P_\pi(Y_{0:n-1})$ and set

$$F_\pi(z) = \Pr(I_\pi \le z), \quad F_*(z) = \min_\pi F_\pi(z), \quad L(T) = \int_0^\infty [1 - F_*(z)]\, dz.$$

Imported theorem. We take as a premise the finite causal Shannon theorem for positive finite controlled trees, in the form

$$\boxed{L(T) \le C(T) \le \mathrm{H}(U_G) \le L(T) + c_\star, \qquad c_\star = \frac{1 + \log_2 e}{2}.} \tag{CS}$$

Here U_G is a finite greedy seed with a causal decoder. The proof and certification of (CS) belong to the companion development [1]; they are not independently audited in this note. All entropies are in bits.

Corollary 1 (One seed for the operational model, conditional on (CS))**.** *Under the finite-alphabet and positivity assumptions, one finite seed realizes the operational model under every deterministic adaptive experimental policy, with the entropy sandwich (CS).*

Proof. The induced tree satisfies the hypotheses of (CS). Apply that theorem and then the exact transcript reduction. □

This is exactness for the specified discretized law. It does not imply exact reproduction of unrecorded physiology, continuous measurements, or interventions outside the prescribed action set.

4 Limits of identification

Proposition 2 (Observable laws determine seed feasibility)**.** *Models with the same experimental interface and the same transcript laws under every deterministic adaptive policy have identical feasible seed–decoder pairs and identical optimal seed entropy.*

Proof. For a fixed seed law and decoder, exactness consists precisely of the equalities between its transcript laws and the target laws. Identical target laws impose identical constraints. □

For example, attach an arbitrary finite hidden coordinate E_t and stochastic kernel $Q_a(e' \mid s, e, s')$. Put

$$\widetilde{K}_a((s,e),(s',e')) = K_a(s,s')Q_a(e' \mid s,e,s'), \qquad \widetilde{O}_a(y \mid s',e') = O_a(y \mid s').$$

Any initial joint law with S-marginal ν produces the same observable laws: summation over e' recovers K_a, and the emission does not depend on e'. Induction gives equivalence under adaptive policies. The seed cost therefore cannot identify this hidden extension.

Silence of reports is not silence of dynamics. If a reported variable R_t is constant, then $\mathrm{H}(R_t) = 0$. Nevertheless, a hidden binary state can be independently refreshed at every step, with $\mathrm{H}(S_1, \ldots, S_n) = n$, or remain constant with zero entropy. Both produce the same constant report. This elementary example, separate from the positivity assumption used for (CS), shows that report entropy alone identifies neither hidden entropy nor energetic cost.

5 A falsifiable program

Compare a model merging all low-content states with one separating preserved responsiveness from impaired access. Fit on training data and evaluate held-out participants and intervention sequences using predictive log scores:

$$\ell(M) = \sum_{\text{held-out records}} \sum_{t} \log p_M(y_t \mid h_t, a_t).$$

Complexity control, uncertainty estimates, and participant-level validation are required. More epochs from one person are not independent replications. Better prediction supports an operational distinction, not unique identification of latent coordinates.

Recent studies of relaxed alertness [2], pure awareness [3], and mind blanking [4] concern distinct experimental contrasts. A revised cessation preprint [5] changed its complexity conclusions after revising its analysis. These findings motivate separating subjective content, responsiveness, and measured dynamics rather than collapsing them into one numerical depth score. EEG complexity measures are not automatically causal seed entropy.

Scope. The model and reduction are mathematically explicit. Their empirical adequacy remains to be tested. Neither (CS) nor the present propositions establish consciousness transfer, physiological closure, vanishing energy cost, or consciousness outside spacetime.

References

[1] J. Mikulík, *Finite Shannon Theorem*, companion repository. `https://github.com/jendamikulik/finite-Shannon-theorem`. A reproducible application should pin and audit the exact source commit.

[2] A. T. Duda, A. R. Clarke, and R. J. Barry (2026). Neural Oscillatory and Sympathetic Arousal Dynamics During Mindfulness Meditation: Characterizing Relaxed Alertness. *Psychophysiology* 63, e70425. https://doi.org/10.1111/psyp.70425.

[3] A. Chandia-Jorquera et al. (2026). Distilling the Neurophenomenological Signatures of Pure Awareness during Transcendental Meditation. *Journal of Cognitive Neuroscience* 38, 2028–2045. https://doi.org/10.1162/JOCN.a.2626.

[4] E. Munoz-Musat et al. (2025). Behavioral, experiential, and physiological signatures of mind blanking. *PNAS* 122, e2510262122. https://doi.org/10.1073/pnas.2510262122.

[5] K. Shinozuka et al. (2026). Neuroelectrophysiological correlates of extended cessation of consciousness in advanced meditators: A multimodal EEG and MEG study. *bioRxiv*, version 3, 17 October 2026. Not peer reviewed. https://www.biorxiv.org/content/10.1101/2025.09.19.677455v3.